

EDEXCEL INTERNATIONAL A LEVEL

# WMA11/01 May/June 2022 Worked Solutions

May/June 2022 paper rebuilt with the worked solution after each question.

**10**

paper questions

**WMA11**

Pure 1

**Single-  
paper  
solutions**standalone IAL  
route**Dr Eslam Ahmed**Prepared for Dr Eslam Ahmed - [eliteigcse.com](https://eliteigcse.com)**P1**

**PAST PAPER**

# **WMA11/01 May/June 2022**

**May/June 2022 | 10 questions | 75 marks**

**10**

questions

**75**

marks

**Answers**

worked solution  
after each  
question

# Question 1

Integration

1. Find

$$\int \left( 10x^5 + 6x^3 - \frac{3}{x^2} \right) dx$$

giving your answer in simplest form.

(4)

(Total 4 marks)

# Worked Solution - Question 1

**1. Rewrite the negative power**

$$\int \left( 10x^5 + 6x^3 - \frac{3}{x^2} \right) dx = \int (10x^5 + 6x^3 - 3x^{-2}) dx.$$

**2. Integrate term by term**

$$\int 10x^5 dx = \frac{10x^6}{6} = \frac{5}{3}x^6, \int 6x^3 dx = \frac{6x^4}{4} = \frac{3}{2}x^4, \text{ and } \int -3x^{-2} dx = 3x^{-1}.$$

**3. Add the constant**

Therefore the integral is  $\frac{5}{3}x^6 + \frac{3}{2}x^4 + \frac{3}{x} + c$ .

**Final answer**

$$\frac{5}{3}x^6 + \frac{3}{2}x^4 + \frac{3}{x} + c.$$

## Question 2

## Basic Trigonometry

2. In the triangle  $ABC$ ,

- $AB = 21$  cm
- $BC = 13$  cm
- angle  $BAC = 25^\circ$
- angle  $ACB = x^\circ$

(a) Use the sine rule to find the value of  $\sin x^\circ$ , giving your answer to 4 decimal places. (2)

Given also that  $AB$  is the longest side of the triangle,

(b) find the value of  $x$ , giving your answer to 2 decimal places. (3)

(Total 5 marks)

## Worked Solution - Question 2

Topic group

**1. Use the sine rule**

The side opposite  $x$  is  $AB = 21$ , and the side opposite  $25^\circ$  is  $BC = 13$ , so

$$\frac{\sin x}{21} = \frac{\sin 25^\circ}{13}.$$

**2. Find the sine value**

$$\sin x = \frac{21 \sin 25^\circ}{13} = 0.68269 \dots$$

**3. Choose the correct angle**

The angle from the calculator is  $43.05^\circ$ , but  $AB$  is the longest side, so  $x$  is obtuse.

Hence  $x = 180^\circ - 43.05^\circ = 136.95^\circ$ .

**Final answer**

$x = 137^\circ$  to 3 significant figures.

## Question 3

3. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

- (i) Show that  $\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}}$  is an integer and find its value.

(2)

- (ii) Simplify

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}}$$

giving your answer in the form  $a + b\sqrt{5}$  where  $a$  and  $b$  are rational numbers.

(3)

(Total 5 marks)

## Worked Solution - Question 3

**1. Simplify the square roots**

$$\sqrt{180} = 6\sqrt{5} \text{ and } \sqrt{80} = 4\sqrt{5}.$$

**2. Complete part (a)**

$$\frac{\sqrt{180} - \sqrt{80}}{\sqrt{5}} = \frac{6\sqrt{5} - 4\sqrt{5}}{\sqrt{5}} = 2.$$

**3. Rationalise part (b)**

$$\frac{4\sqrt{5} - 5}{7 - 3\sqrt{5}} \times \frac{7 + 3\sqrt{5}}{7 + 3\sqrt{5}} = \frac{(4\sqrt{5} - 5)(7 + 3\sqrt{5})}{49 - 45}.$$

**4. Expand and simplify**

The numerator is  $28\sqrt{5} + 60 - 35 - 15\sqrt{5} = 25 + 13\sqrt{5}$ , so the expression is  $\frac{25 + 13\sqrt{5}}{4}$ .

**Final answer**

(a) 2.

(b)  $\frac{25}{4} + \frac{13}{4}\sqrt{5}$ .



## Question 4

## Transformations

4.

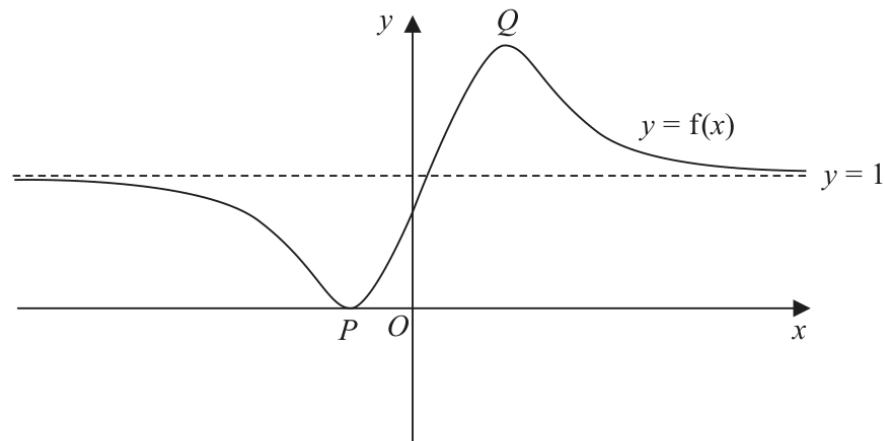


Figure 1

Figure 1 shows a sketch of a curve with equation  $y = f(x)$

The curve has a minimum at  $P(-1, 0)$  and a maximum at  $Q\left(\frac{3}{2}, 2\right)$

The line with equation  $y = 1$  is the only asymptote to the curve.

On separate diagrams sketch the curves with equation

(i)  $y = f(x) - 2$  (3)

(ii)  $y = f(-x)$  (3)

On each sketch you must clearly state

- the coordinates of the maximum and minimum points
- the equation of the asymptote

(Total 6 marks)

# Worked Solution - Question 4

**1. Translate for  $y = f(x) - 2$** 

Subtracting 2 moves every point of the graph down by 2 units.

**2. State the new key features**

The minimum  $P(-1, 0)$  becomes  $(-1, -2)$ , the maximum  $Q(\frac{3}{2}, 2)$  becomes  $(\frac{3}{2}, 0)$ , and the asymptote  $y = 1$  becomes  $y = -1$ .

**3. Reflect for  $y = f(-x)$** 

Replacing  $x$  by  $-x$  reflects the graph in the y-axis.

**4. State the reflected key features**

The minimum  $P(-1, 0)$  becomes  $(1, 0)$ , the maximum  $Q(\frac{3}{2}, 2)$  becomes  $(-\frac{3}{2}, 2)$ , and the horizontal asymptote remains  $y = 1$ .

**Final answer**

(i) minimum  $(-1, -2)$ , maximum  $(\frac{3}{2}, 0)$ , asymptote  $y = -1$ . (ii) minimum  $(1, 0)$ , maximum  $(-\frac{3}{2}, 2)$ , asymptote  $y = 1$ .

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9 marks

## Question 5

Straight Line

5. The curve  $C$  has equation  $y = f(x)$

Given that

- $f(x)$  is a quadratic expression
- the maximum turning point on  $C$  has coordinates  $(-2, 12)$
- $C$  cuts the negative  $x$ -axis at  $-5$

- (a) find  $f(x)$

(4)

The line  $l_1$  has equation  $y = \frac{4}{5}x$

Given that the line  $l_2$  is perpendicular to  $l_1$  and passes through  $(-5, 0)$

- (b) find an equation for  $l_2$ , writing your answer in the form  $y = mx + c$  where  $m$  and  $c$  are constants to be found.

(3)

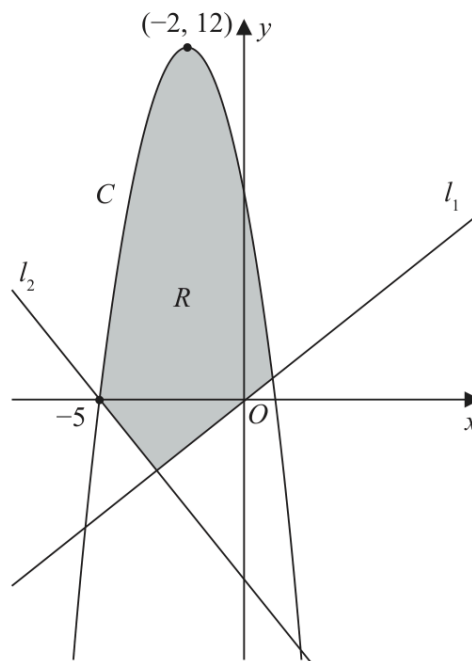


Figure 2

Figure 2 shows a sketch of the curve  $C$  and the lines  $l_1$  and  $l_2$

- (c) Define the region  $R$ , shown shaded in Figure 2, using inequalities.

(2)

(Total 9 marks)

## Worked Solution - Question 5

**1. Use symmetry of the quadratic**

The maximum is at  $x = -2$ , and one root is  $x = -5$ . The other root is the same distance to the right of  $-2$ , so it is  $x = 1$ .

**2. Find the quadratic**

Let  $f(x) = a(x + 5)(x - 1)$ . Since the maximum point is  $(-2, 12)$ ,  
 $12 = a(3)(-3) = -9a$ , so  $a = -\frac{4}{3}$ .

**3. Find the perpendicular gradient**

The gradient of  $l_1$  is  $\frac{4}{5}$ , so the gradient of a perpendicular line is  $-\frac{5}{4}$ .

**4. Find  $l_2$** 

Using the point  $(-5, 0)$ ,  $y = -\frac{5}{4}x + c$ . Then  $0 = \frac{25}{4} + c$ , so  $c = -\frac{25}{4}$  and  $l_2$  is  
 $y = -\frac{5}{4}x - \frac{25}{4}$ .

**5. Write the inequalities for R**

The region is below the curve and above both straight lines, so  $y \leq f(x)$ ,  
 $y \geq \frac{4}{5}x$ , and  $y \geq -\frac{5}{4}x - \frac{25}{4}$ .

**Final answer**

$$(a) f(x) = -\frac{4}{3}(x + 5)(x - 1).$$

$$(b) y = -\frac{5}{4}x - \frac{25}{4}.$$

$$(c) y \leq f(x), y \geq \frac{4}{5}x, y \geq -\frac{5}{4}x - \frac{25}{4}.$$

## Question 6

## Quadratics

6. In this question you must show all stages of your working.  
Solutions relying on calculator technology are not acceptable.

- (a) Given that

$$2xy - 3x^2 = 50$$

and

$$y - x^3 + 6x = 0$$

show that

$$2x^4 - 15x^2 - 50 = 0 \quad (2)$$

- (b) Hence solve the simultaneous equations

$$2xy - 3x^2 = 50$$

$$y - x^3 + 6x = 0$$

Give your answers in fully simplified surd form.

(5)

(Total 7 marks)

## Worked Solution - Question 6

**1. Rearrange the second equation**

From  $y - x^3 + 6x = 0$ , we get  $y = x^3 - 6x$ .

**2. Substitute into the first equation**

$2x(x^3 - 6x) - 3x^2 = 50$ , so  $2x^4 - 15x^2 - 50 = 0$ .

**3. Solve the quadratic in  $x$  squared**

Let  $u = x^2$ . Then  $2u^2 - 15u - 50 = 0$ , giving  $u = 10$  or  $u = -\frac{5}{2}$ .

**4. Reject the negative value**

Since  $x^2$  cannot be negative,  $x^2 = 10$ , so  $x = \pm\sqrt{10}$ .

**5. Find  $y$** 

$y = x^3 - 6x = x(x^2 - 6) = x(10 - 6) = 4x$ . Therefore  $y = 4\sqrt{10}$  when  $x = \sqrt{10}$  and  $y = -4\sqrt{10}$  when  $x = -\sqrt{10}$ .

**Final answer**

$(x, y) = (\sqrt{10}, 4\sqrt{10})$  or  $(-\sqrt{10}, -4\sqrt{10})$ .

## Question 7

## Integration

7. The curve  $C$  has equation  $y = f(x)$ ,  $x > 0$

Given that

- $f'(x) = \frac{2}{\sqrt{x}} + \frac{A}{x^2} + 3$ , where  $A$  is a constant
- $f''(x) = 0$  when  $x = 4$

(a) find the value of  $A$ .

(4)

Given also that

- $f(x) = 8\sqrt{3}$ , when  $x = 12$

(b) find  $f(x)$ , giving each term in simplest form.

(5)

(Total 9 marks)



## Worked Solution - Question 7

**1. Differentiate  $f'(x)$** 

$$f'(x) = 2x^{-1/2} + Ax^{-2} + 3, \text{ so } f''(x) = -x^{-3/2} - 2Ax^{-3}.$$

**2. Use  $f''(4) = 0$** 

$$-4^{-3/2} - 2A(4^{-3}) = 0, \text{ so } -\frac{1}{8} - \frac{A}{32} = 0 \text{ and } A = -4.$$

**3. Substitute  $A$  into  $f'(x)$** 

$$f'(x) = \frac{2}{\sqrt{x}} - \frac{4}{x^2} + 3.$$

**4. Integrate to find  $f(x)$** 

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x + c.$$

**5. Use  $f(12) = 8\sqrt{3}$** 

$$f(12) = 8\sqrt{3} + \frac{1}{3} + 36 + c = 8\sqrt{3}, \text{ so } c = -\frac{109}{3}.$$

**Final answer**

$$f(x) = 4\sqrt{x} + \frac{4}{x} + 3x - \frac{109}{3}.$$

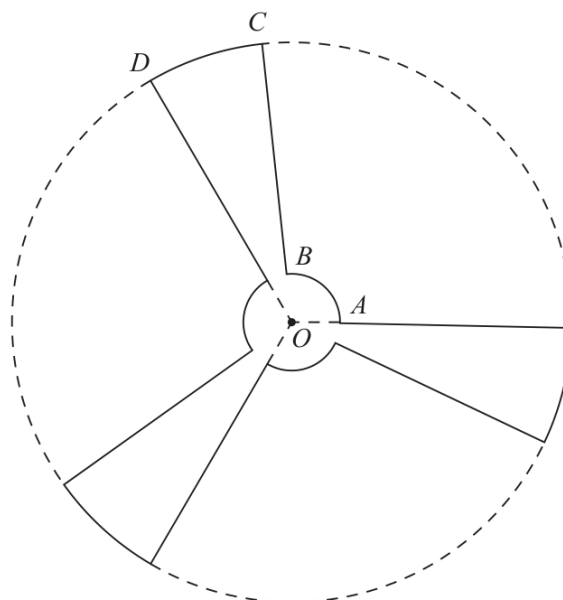
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10 marks

## Question 8

Radians

8.



**Figure 3**

Figure 3 shows a sketch of the outline of the face of a ceiling fan viewed from below.

The fan consists of three identical sections congruent to  $OABCD$ , shown in Figure 3, where

- $OABO$  is a sector of a circle with centre  $O$  and radius 9 cm
- $OBCDO$  is a sector of a circle with centre  $O$  and radius 84 cm
- angle  $AOD = \frac{2\pi}{3}$  radians

Given that the length of the arc  $AB$  is 15 cm,

- (a) show that the length of the arc  $CD$  is 35.9 cm to one decimal place. (3)

The face of the fan is modelled to be a flat surface.

Find, according to the model,

- (b) the perimeter of the face of the fan, giving your answer to the nearest cm, (2)
- (c) the surface area of the face of the fan.

Give your answer to 3 significant figures and make your units clear. (5)

**(Total 10 marks)**

## Worked Solution - Question 8

**1. Find the small sector angle**

For arc  $AB$ ,  $s = r\theta$ , so  $15 = 9\theta$  and  $\theta = \frac{5}{3}$  radians.

**2. Find the outer arc angle**

Each identical section has total angle  $\frac{2\pi}{3}$ , so the angle for arc  $CD$  is

$$\frac{2\pi}{3} - \frac{5}{3} = \frac{2\pi - 5}{3}.$$
**3. Find  $CD$** 

$$CD = 84 \left( \frac{2\pi - 5}{3} \right) = 28(2\pi - 5) = 35.9 \text{ cm to 3 significant figures.}$$

**4. Find the perimeter**

There are 3 inner arcs, 3 outer arcs, and 6 straight radial edges of length  $84 - 9 = 75$ . Thus  $P = 3(15) + 3(35.929\dots) + 6(75) = 602.8\dots \text{ cm}$ .

**5. Find the area of one section**

One section has area  $\frac{1}{2}(9^2) \left( \frac{5}{3} \right) + \frac{1}{2}(84^2) \left( \frac{2\pi - 5}{3} \right) = 1576.525\dots \text{ cm}^2$ .

**6. Find the total area**

The total area is  $3(1576.525\dots) = 4729.57\dots \text{ cm}^2 = 4.73 \times 10^3 \text{ cm}^2$  to 3 significant figures.

**Final answer**

(a)  $CD = 35.9 \text{ cm.}$

(b)  $603 \text{ cm.}$

(c)  $4.73 \times 10^3 \text{ cm}^2.$

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8 marks

## Question 9

Trigonometric Functions

9.

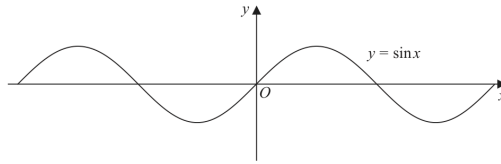


Figure 4

Figure 4 shows part of the graph of the curve with equation  $y = \sin x$

Given that  $\sin \alpha = p$ , where  $0 < \alpha < 90^\circ$

- (a) state, in terms of  $p$ , the value of

- (i)  $2 \sin(180^\circ - \alpha)$   
(ii)  $\sin(\alpha - 180^\circ)$   
(iii)  $3 + \sin(180^\circ + \alpha)$

(3)

A copy of Figure 4, labelled Diagram 1, is shown on page 27.

On Diagram 1,

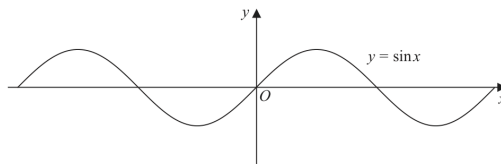
- (b) sketch the graph of  $y = \sin 2x$

(2)

- (c) Hence find, in terms of  $\alpha$ , the  $x$  coordinates of any points in the interval  $0 < x < 180^\circ$  where

$$\sin 2x = p$$

(3)



**Diagram 1**

**(Total 8 marks)**

# Worked Solution - Question 9

**1. Use sine symmetry**

$$\sin(180^\circ - \alpha) = \sin \alpha = p, \text{ so } 2 \sin(180^\circ - \alpha) = 2p.$$

**2. Use odd symmetry**

$$\sin(\alpha - 180^\circ) = -\sin(180^\circ - \alpha) = -p.$$

**3. Use the third quadrant identity**

$$\sin(180^\circ + \alpha) = -\sin \alpha = -p, \text{ so } 3 + \sin(180^\circ + \alpha) = 3 - p.$$

**4. Sketch  $y = \sin 2x$** 

The graph has amplitude 1 and period  $180^\circ$ . It crosses the x-axis at  $0^\circ, 90^\circ, 180^\circ$ , has a maximum at  $45^\circ$ , and a minimum at  $135^\circ$ .

**5. Solve  $\sin 2x = p$** 

Since  $\sin \alpha = p$  and  $0 < 2x < 360^\circ$ , the possible values are  $2x = \alpha$  or  $2x = 180^\circ - \alpha$ .

**6. Find  $x$** 

$$\text{Therefore } x = \frac{\alpha}{2} \text{ or } x = 90^\circ - \frac{\alpha}{2}.$$

**Final answer**

$$(a)(i) 2p, (ii) -p, (iii) 3 - p.$$

$$(c) x = \frac{\alpha}{2}, 90^\circ - \frac{\alpha}{2}.$$



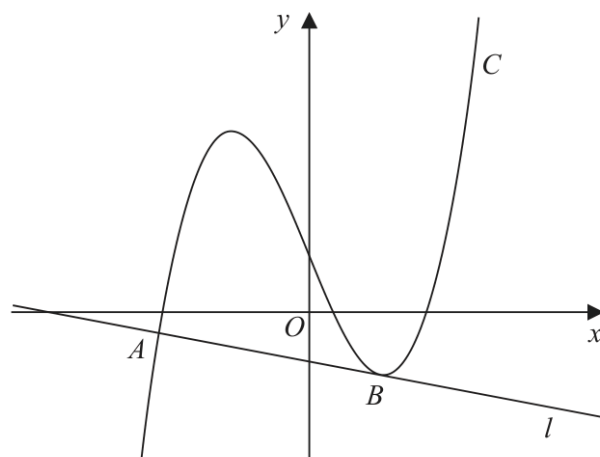
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12 marks

## Question 10

Differentiation

10.



**Figure 5**

Figure 5 shows a sketch of the curve  $C$  with equation

$$y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$$

where  $k$  is a constant.

- (a) Find  $\frac{dy}{dx}$  (2)

The line  $l$ , shown in Figure 5, is the normal to  $C$  at the point  $A$  with  $x$  coordinate  $-\frac{7}{2}$

Given that  $l$  is also a tangent to  $C$  at the point  $B$ ,

- (b) show that the  $x$  coordinate of the point  $B$  is a solution of the equation

$$12x^2 + 4x - 33 = 0$$

(4)

- (c) Hence find the  $x$  coordinate of  $B$ , justifying your answer. (2)

Given that the  $y$  intercept of  $l$  is  $-1$

- (d) find the value of  $k$ . (4)

**(Total 12 marks)**

## Worked Solution - Question 10

**1. Differentiate the curve**

For  $y = \frac{2}{7}x^3 + \frac{1}{7}x^2 - \frac{5}{2}x + k$ ,  $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}$ .

**2. Find the tangent gradient at A**

At  $x = -\frac{7}{2}$ ,  $\frac{dy}{dx} = \frac{6}{7}\left(\frac{49}{4}\right) + \frac{2}{7}\left(-\frac{7}{2}\right) - \frac{5}{2} = \frac{21}{2} - 1 - \frac{5}{2} = 7$ .

**3. Find the gradient of l**

Since  $l$  is the normal at  $A$ , its gradient is the negative reciprocal of 7, so

$$m_l = -\frac{1}{7}.$$

**4. Use that l is tangent at B**

At  $B$ , the tangent gradient is also  $-\frac{1}{7}$ , so  $\frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2} = -\frac{1}{7}$ .

**5. Reach the given quadratic**

Multiplying by 14 gives  $12x^2 + 4x - 35 = -2$ , hence  $12x^2 + 4x - 33 = 0$ .

**6. Solve for the x-coordinate of B**

$12x^2 + 4x - 33 = 0$  gives  $x = \frac{-4 \pm 40}{24}$ , so  $x = \frac{3}{2}$  or  $x = -\frac{11}{6}$ . From the diagram,  $B$  is the point to the right of the y-axis, so  $x_B = \frac{3}{2}$ .

**7. Use the y-intercept of l**

Since the y-intercept is  $-1$  and the gradient is  $-\frac{1}{7}$ , the line is  $y = -\frac{1}{7}x - 1$ .

**8. Find the y-coordinate of A**

At  $x = -\frac{7}{2}$  on the line,  $y = -\frac{1}{7}\left(-\frac{7}{2}\right) - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$ .

**9. Substitute A into the curve**

$$-\frac{1}{2} = \frac{2}{7} \left(-\frac{7}{2}\right)^3 + \frac{1}{7} \left(-\frac{7}{2}\right)^2 - \frac{5}{2} \left(-\frac{7}{2}\right) + k = -\frac{7}{4} + k.$$

**10. Find k**

Therefore  $k = \frac{5}{4}$ .

**Final answer**

(a)  $\frac{dy}{dx} = \frac{6}{7}x^2 + \frac{2}{7}x - \frac{5}{2}.$

(c)  $x_B = \frac{3}{2}.$

(d)  $k = \frac{5}{4}.$